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URBAN CANINE SAFETY BELT AND MONITORING SYSTEM

WHERE TECHNOLOGY MEETS COMPASSION

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DOMAIN:INTERNET OF THINGS

The Urban Canine Safety Belt mainly comes under the **Internet of Things(IoT)** domain because it uses sensors and connected devices to monitor the dog's health and send real-time data. It also involves **embedded systems** like ESP32 and is related to healthcare technology as it focuses on animal welfare technology,specifically aimed at improving the safety and health of street dogs.

ABSTRACT:

The Urban Canine Safety Belt is designed to reduce dog bite incidents in urban areas through a humane approach. Aggressive situations are inferred using predefined physical triggers and a **non-lethal ultrasonic** sound is activated to interrupt aggressive behavior. The system provides real-time intervention while ensuring ethical animal management and safer coexistence between humans and street dogs.

INTRODUCTION:

Street dogs are common in many areas and often live without proper health monitoring, which can lead to diseases like rabies and increase dog bite cases. This creates a risk to both humans and animals. To address this, a smart system can be used to detect early health issues in dogs. This project also focuses on reducing dog bites using an ultrasonic mechanism that helps to distract or repel dogs during aggressive behavior, improving both public safety and animal welfare.

Problem Statement:

- Street dogs are not regularly monitored, which leads to the spread of diseases like rabies and increases dog bite cases, creating a public safety .
- The main aim of this project is to reduce dog bites using an ultrasonic mechanism behavior to distract or repel the dog.
- This project is chosen to improve both human safety and animal welfare, as many dogs remain untreated while people are at risk.
- According to the World Health Organization, dog bites causes about 59,000 deaths per year globally, and India reports nearly 1.5–2 crore dog bite cases annually, especially during 2022–2025.
- Both rural and urban areas are affected, but rural areas are more affected due to lack of healthcare and vaccination, while urban areas face issues due to high population and interaction.

THE PROBLEM:



Literature Survey				
S. No	Title of the Paper	Author	Year	Drawbacks
1	Wearable Sensors-Based Intelligent Animal Monitoring	Ding et al.	2025	High computational cost; reduced real-time accuracy; not suitable for complex environments
2	IoT-Based Livestock Health Surveillance Systems	Modak et al.	2025	Needs continuous internet; high power consumption; privacy issues
3	Wearable Biosensors for Animal Health Monitoring	Krishnendu & Singh	2025	High cost; limited availability; requires calibration
4	Intelligent Wearable Device for Real-Time Cattle Health Monitoring	Yu et al.	2024	Designed for cattle; depends on sensor placement; limited real-world testing
5	IoT-Based Animal Health Monitoring Smart Collar	Maharajpet et al.	2024	Limited prediction accuracy; basic monitoring; battery dependency

PROPOSED SOLUTION

KEY FEATURES:

BEHAVIOUR DETECTION:Sensors (motion+ultrasonic) to identify aggression.

HARMLESS DETERRENTS:

Vibrations,sound alerts,pheromone spray.

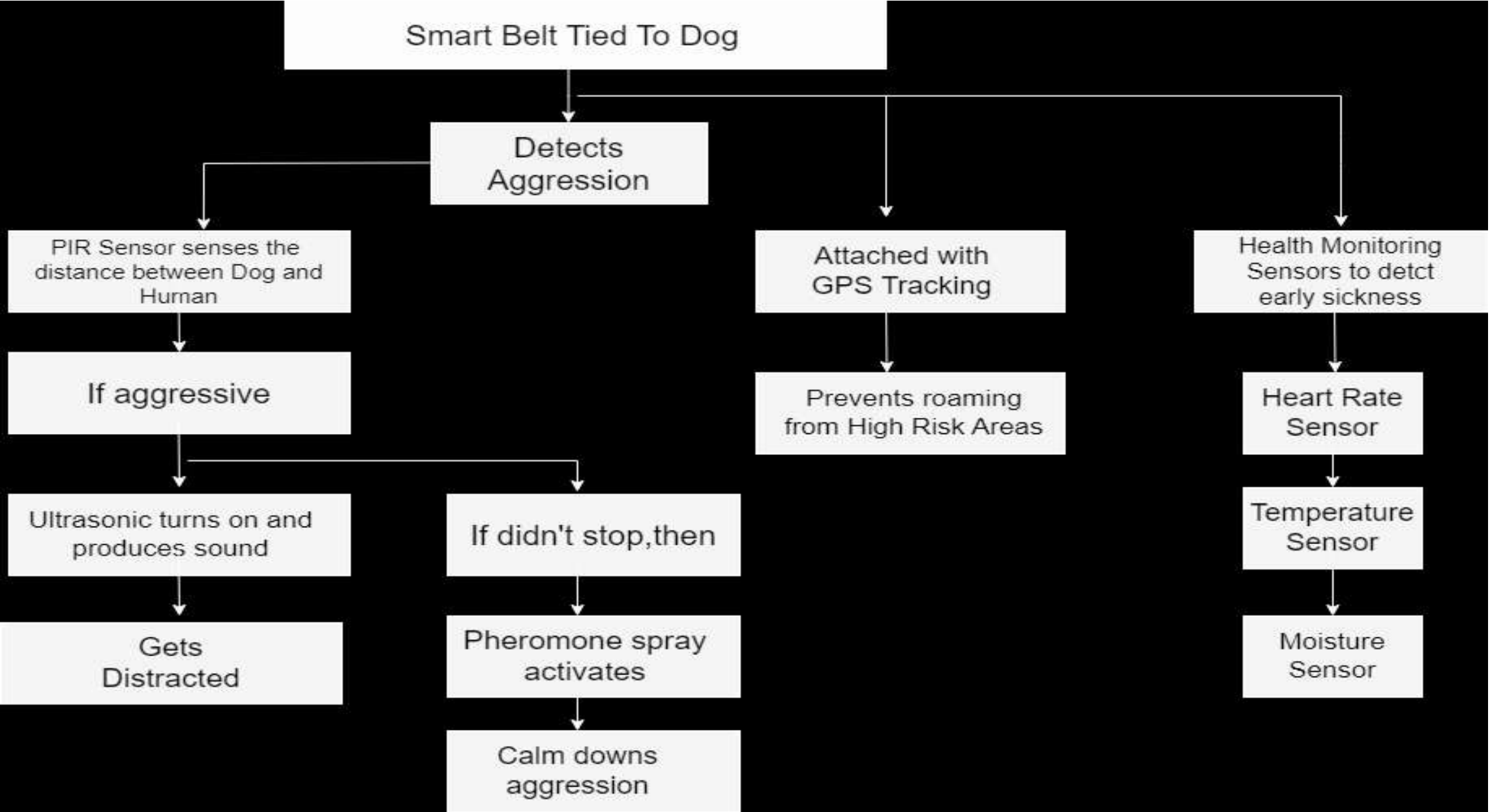
GPS TRACKING:Monitor dog Movement , prevent roaming in High-risk areas.

HEALTH MONITORING:Detect early signs of rabies or sickness.

BLUETOOTH INTEGRATION: Enables real-time monitoring of sensor data and alerts through a mobile application.



SYSTEM ARCHITECTURE



OUTCOME AND BENEFITS:

COMMUNITY BENEFITS:

- Better coexistence between humans and street dogs.
- Reduction in public fear and complaints.

COST & RESOURCE EFFICIENT:

- Saves healthcare costs by reducing bite-related treatments.
- Cuts municipal expenses on dog conflict management.

SCALABILITY & INNOVATION:

- Can be adopted in multiple cities.
- Creates opportunities for collaboration with NGOs, pet companies and government.



OUR IMPACT:



IMPROVED PUBLIC SAFETY

Significant reduction in dog bite cases

Safer streets for Children , Pedestrians and Commuters

HUMANE ANIMAL MANAGEMENT

Non-lethal,cruelty-free deterrents

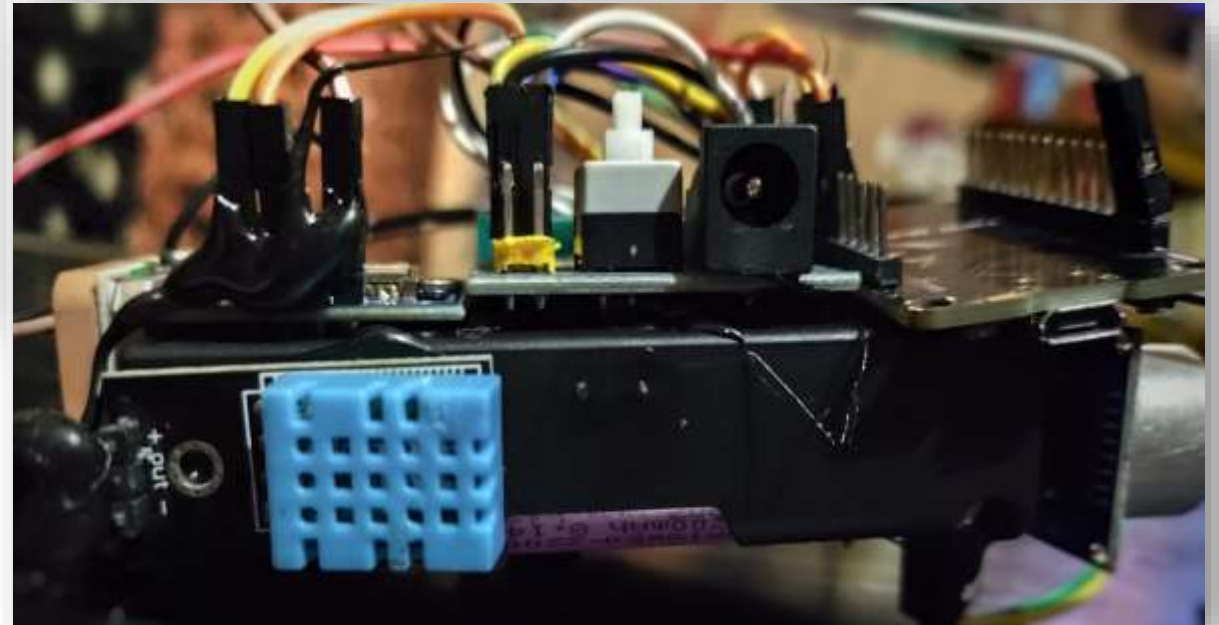
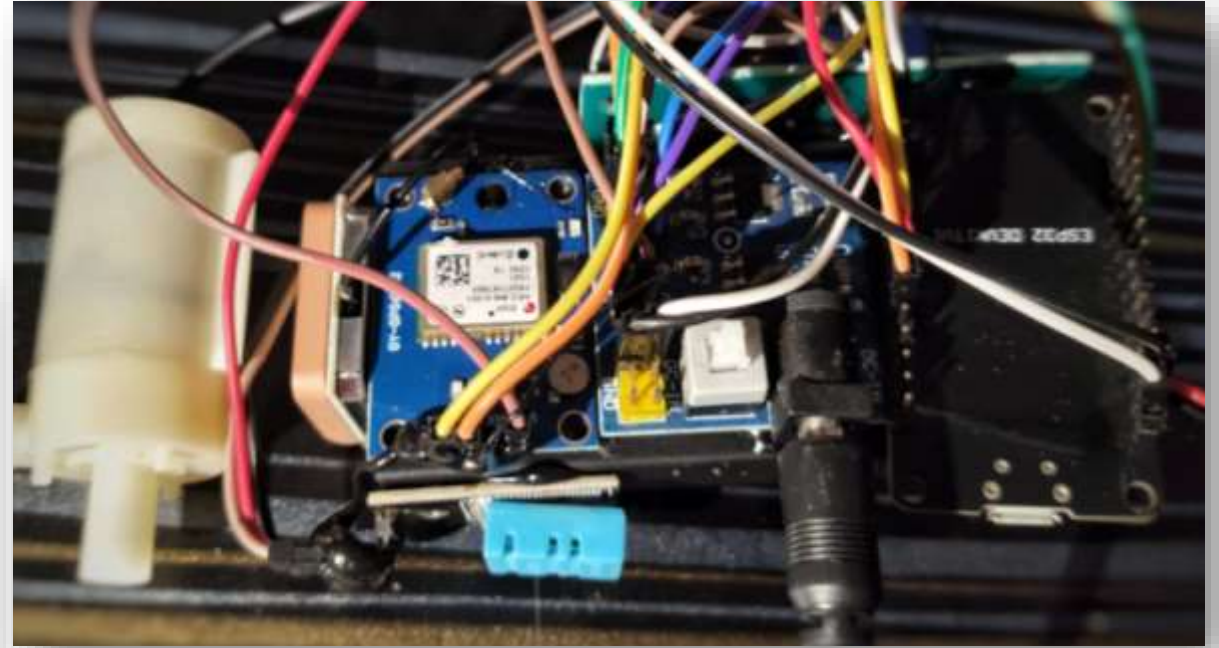
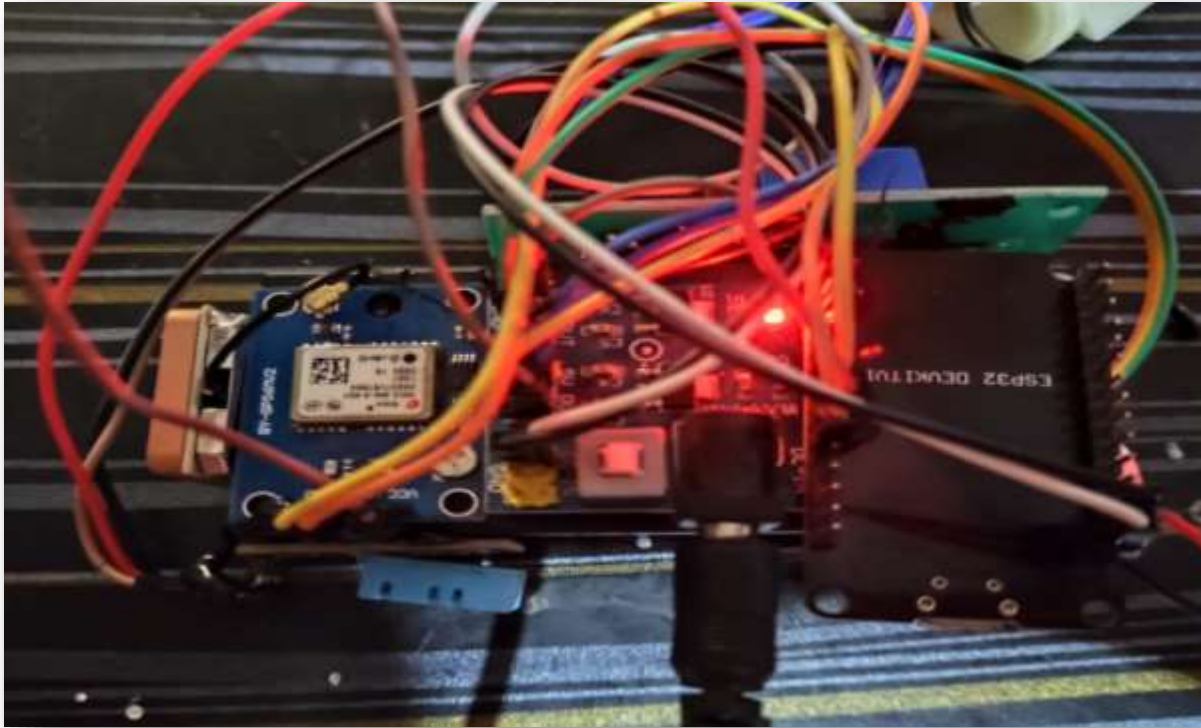
Supports NGO and government ethics.

MEASURABLE SOCIAL & BUSINESS IMPACT

Trackable KPIs : belts depoyed ,bite reduction, Safer streets for children, Supports NGO and adoption rate.

Builds a strong humane-Tech brand presence.

OUTPUT AND RESULT :



CONCLUSION:

This project presents a Smart Street Dog Safety Belt that combines health monitoring and ultrasonic bite prevention to improve public safety and animal welfare. It helps reduce dog bite incidents and supports early detection of health issues, providing a simple and effective solution. A step towards safer streets , smarter cities and a kinder tomorrow.

REFERENCES

- 1.Ding et al., “Wearable Sensors-Based Intelligent Animal Monitoring,” Sensors Journal, 2025.
- 2.Modak et al., “IoT-Based Livestock Health Surveillance Systems,” IET Wireless Sensor Systems, 2025.
- 3.Krishnendu & Singh, “Wearable Biosensors for Animal Health Monitoring,” Elsevier, 2025.
- 4.Yu et al., “Intelligent Wearable Device for Real-Time Cattle Health Monitoring,” Frontiers in Robotics & AI, 2024.
- 5.Maharajpet et al., “IoT-Based Animal Health Monitoring Smart Collar,” EAS Journal, 2024.



Thank you

